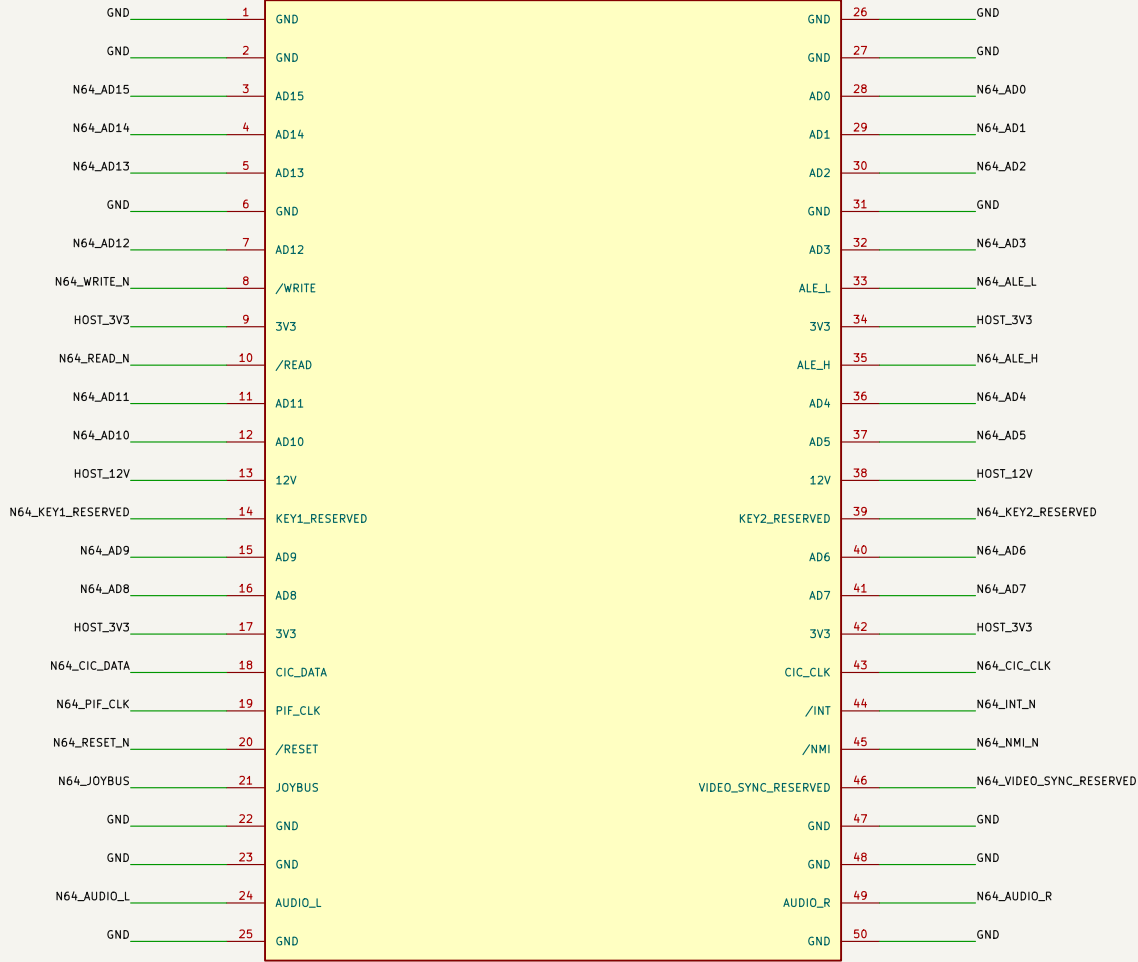


N64 / M64 HOST EDGE

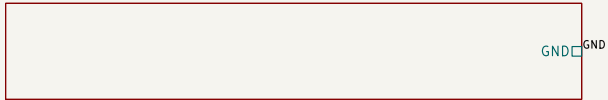
Same 50-contact interface for both host systems

J1
N64_Cartridge_Edge_50



HOST INTERFACE
3.3 V logic; power inputs remain separate from local rails.
AD[15:0] is bidirectional. CIC_DATA / JOYBUS require release.
Reserved contacts and audio are identified only; no drivers fitted.
HOST_12V is reserved, with no selected load. No 5 V edge supply.

USB-C programmer



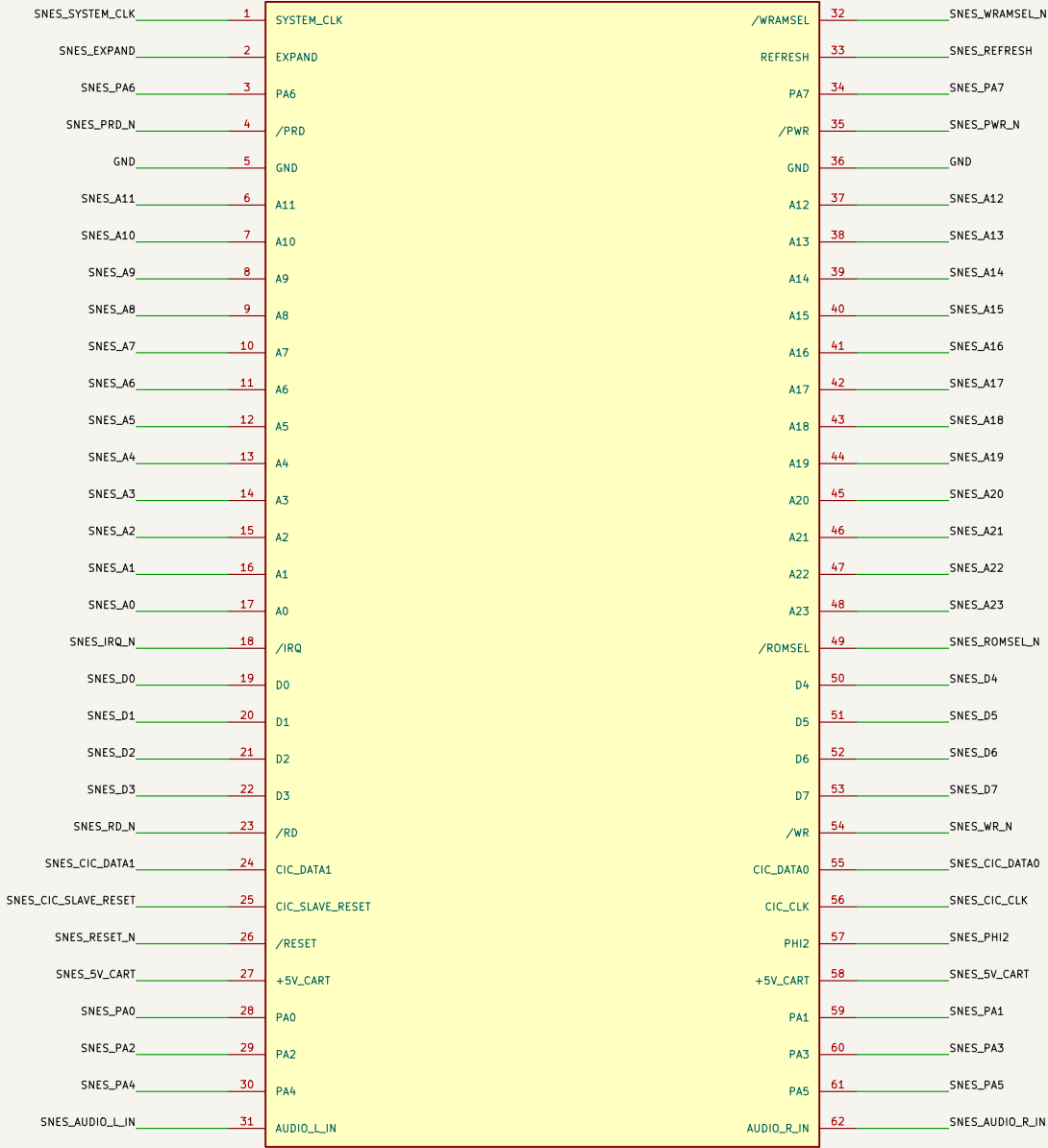
File: usb-programmer.kicad_sch

DRAFT BOUNDARY: cartridge connectors + USB programmer child sheet. Main FPGA, system power, cartridge translation and A/V remain pending.
Only GND is common between the two connectors. Matching names within one domain identify the same net; _N means active low.
Connector pin types are passive; electrical directions and source evidence are in interfaces/*.csv and docs/design/*-interface-notes.md.

SNES / SUPER FAMICOM SOCKET

All 62 contacts retained, including analog cartridge audio

J2
SNES_Female_Slot_62



SNES INTERFACE
5 V cartridge logic requires reviewed translation and isolation.
SNES_5V_CART: protected switched supply, default OFF (pending).
System RESET must support cartridge assertion and input sensing.
CIC_SLAVE_RESET is separate; AUDIO_L/R_IN are analog inputs.
No universal NC contacts. Full pin behavior is recorded in the CSV.

Source attribution and limits: hardware/sn64/README.md
INTERFACE DRAFT ONLY – NOT FOR MANUFACTURE

SN 64

Sheet: /
File: sn64.kicad_sch

Title: SN 64 – cartridge interfaces

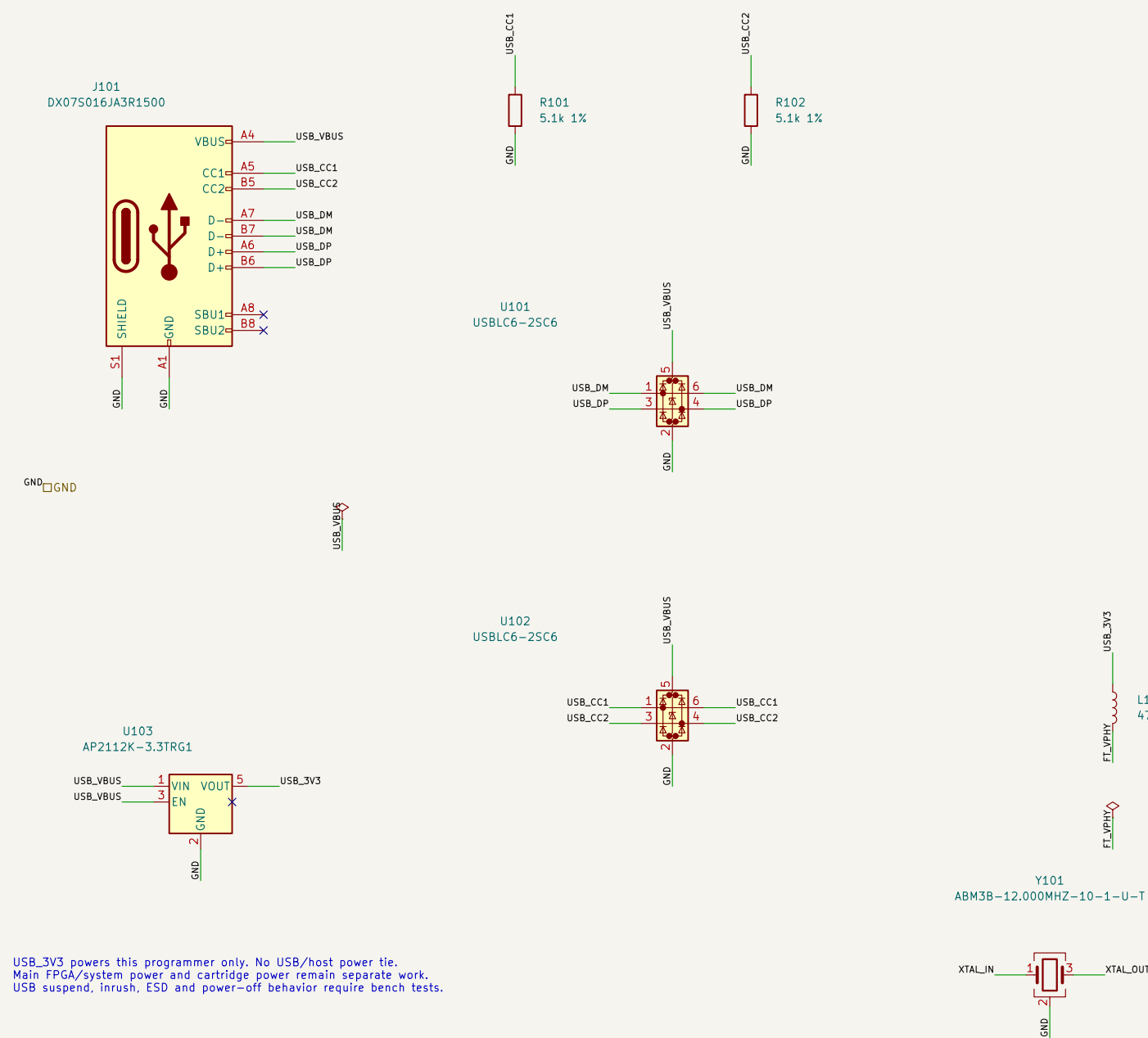
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-28

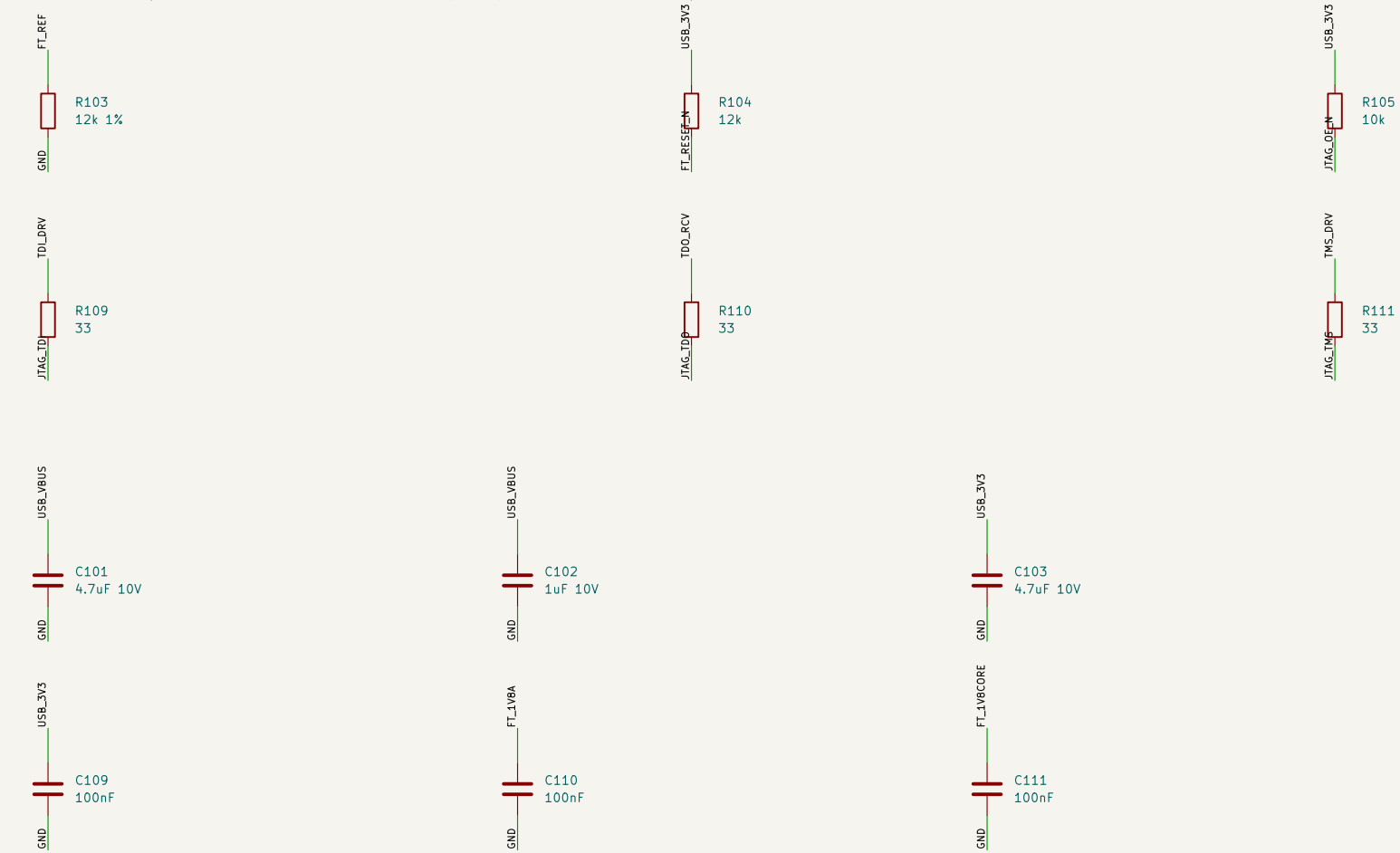
Rev: 0.2-usb
Id: 1/2

SIDE USB-C: INITIAL PROGRAMMING + RECOVERY

USB 2.0 device; 5V input only. Side location is mandatory; exact PCB/enclosure placement remains open.

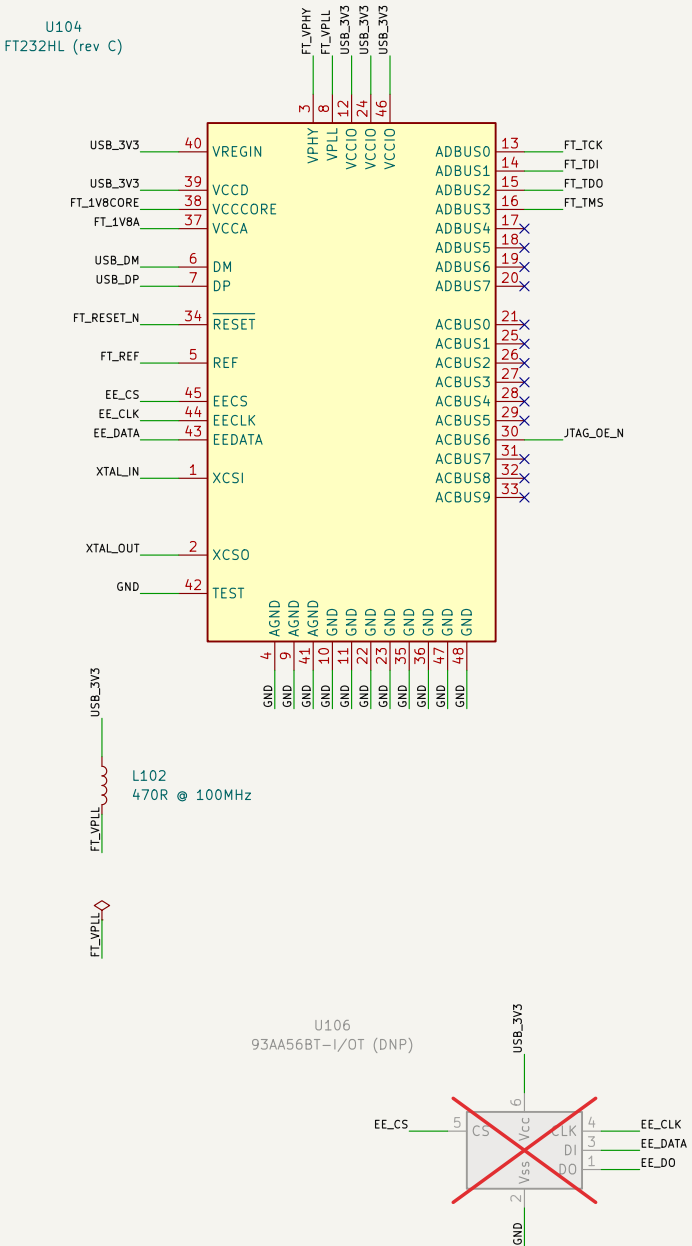


SUPPORT PASSIVES (place at their associated IC/pin; electrical grouping here is not PCB placement)



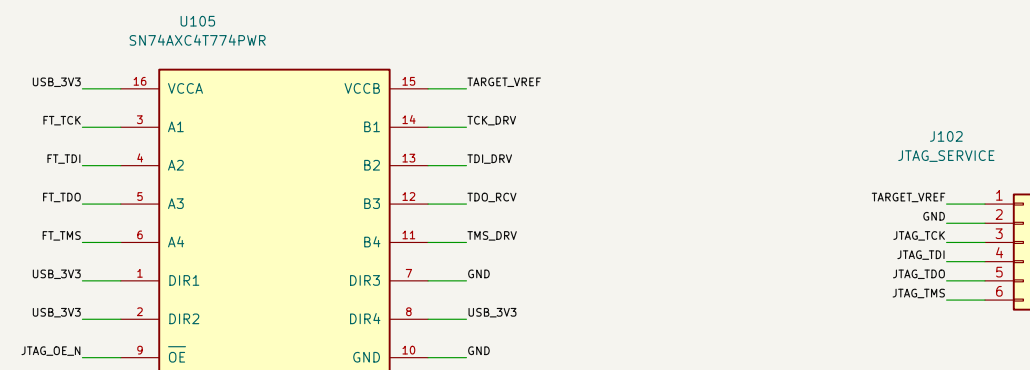
DRAFT 0.2: USB programmer circuitry only. FPGA/storage choice, persistent-image map, system power, routing and physical validation are pending.
Sources: SummerCart64 a1e7996 (CERN-OHL-S-2.0), FTDI FT232HL v2.2, TI SN74AXC47774, JAE drawing SJ122205. See docs/design/usb-programming-architecture.md.

FT232H HARDWARE BRIDGE



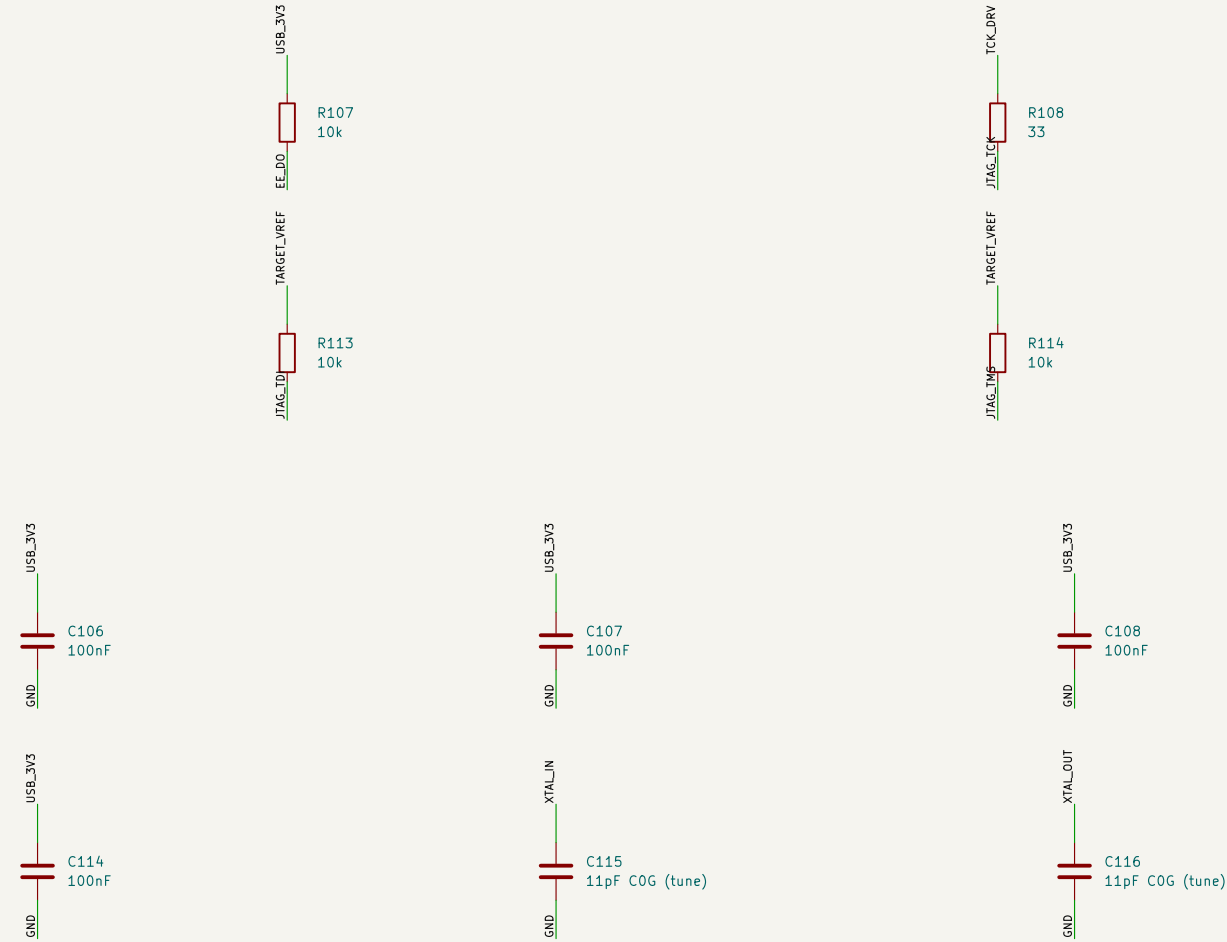
12MHz crystal replaces SC64 external oscillator.
11pF caps are provisional; verify frequency/load after layout.
Optional blank EEPROM: no main-FPGA firmware needed to enumerate.

TARGET ISOLATION + SERVICE ACCESS



TARGET_VREF comes from the future FPGA programming rail.
It powers the B-side translator; it is NOT a USB power output.
JTAG_TCK/TDI/TDO/TMS need final FPGA/flash integration.
Only one programmer may own JTAG at a time.

DEFAULT DISABLED: R105 pulls /OE high. FTDI ACBUS6 is input/pulled-up at blank startup. Host enters MPSSE first, then enables the buffer with --status-pin 14. Generic ft232 without that flag leaves the buffer disabled.



DESIGN DRAFT – NOT FOR MANUFACTURE

SN 64

Sheet: /USB-C programmer/
File: usb-programmer.kicad_sch

Title: SN 64 – side USB-C programmer

Size: A2	Date: 2026-09-29
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Rev: 0.2-usb

Id: 2/2